Sahaj Bhadja

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Summary

Infrastructure-oriented software engineer building **automation, monitoring, diagnostic, and management solutions** with Python, React, TypeScript, Ansible, Shell, Linux, Docker, Kubernetes, LLMs, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Developed robust automated Python services for data workflows, monitoring execution, and troubleshooting failures, processing 5,000+ monthly jobs with validation, reconciliation, and 99%+ reliability across cloud-backed production platforms.
- Analyzed application and database performance for 500+ users, reducing latency from 1.2 seconds to 350 milliseconds through benchmark-driven query optimization, indexing, profiling, and permanent reliability improvements at scale.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Built Python backend services and high-performance React interfaces on AWS, serving 150+ users and 10,000+ daily requests while maintaining stable production operation through automated testing and monitoring.
- Evaluated LLM-assisted approaches using Claude, OpenAI, and Qwen, implementing concurrent processing, model fallbacks, and diagnostic logging while reducing workflow turnaround 60% across 10,000+ submissions and improving operational consistency.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Configured and analyzed 10,000+ performance benchmark workloads across 100,000+ trajectory records, improving long-horizon planning 18% through Python automation, reproducible simulation, algorithm optimization, and reliable evaluation at scale.
- Developed performance analysis infrastructure with PyTorch, Hydra, and experiment tracking, comparing multiple architectures and improving closed-loop stability 15% through systematic validation, debugging, and technical documentation of results.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Built cross-platform management and diagnostic automation using Python, Java, Shell, Jenkins, and virtual machines, executing 20+ tests across Linux, Windows, and macOS while reliably improving pipeline efficiency 68%.
- Troubleshot software, automation, and integration issues using AWS Glue and S3, reducing response times 30%, payload sizes 25%, and technical debt 20% through maintainable permanent solutions at scale.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Automated infrastructure provisioning and configuration using ARM templates, Docker, Jenkins, and Git, reliably supporting eight modular services and two million synchronized records while reducing deployment time 50%.
- Implemented monitoring and diagnostics with Sentry, Azure Monitor, Kibana, and Google Analytics, identifying performance issues and operational improvements that reduced resource waste 42% and increased conversions 15%.

Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, Shell, Bash
Frontend: React, Next.js, HTML, CSS, TailwindCSS, responsive UI, componentization
Infrastructure: Ansible, Linux, Unix, Docker, Kubernetes, Terraform, virtual machines, ARM templates
Operations: Provisioning, configuration, monitoring, diagnostics, troubleshooting, performance analysis, benchmarking
Engineering: Git, Jenkins, CI/CD, automated testing, REST APIs, LLMs, distributed systems, technical documentation

Projects

Linguistix | *Python, AWS, Automated Deployment*

- Developed a containerized Python service with automated validation and deployment workflows on AWS, achieving 87% measured accuracy and dependable operation through repeatable CI/CD-based infrastructure and release checks.

Chat2SQL Code | *Java, TypeScript, Diagnostic Tools*

- Built an AI-assisted database management workspace supporting 10+ platforms, implementing configuration, schema diagnostics, query validation, operational documentation, and cross-platform debugging workflows through integrated Java and TypeScript components.

CPU-scheduling Simulator Live | *C++, Performance Benchmarks*

- Implemented six scheduling algorithms with automated performance metrics and Gantt visualization, analyzing 1,000+ benchmark workloads while reducing average turnaround time 25% and debugging time 20% through systematic testing.